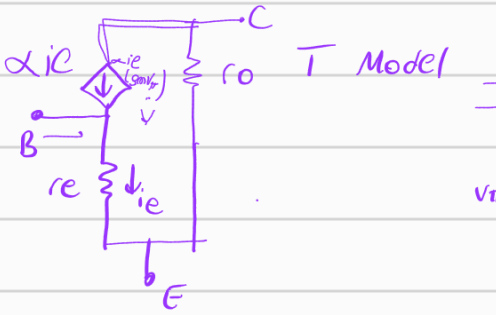
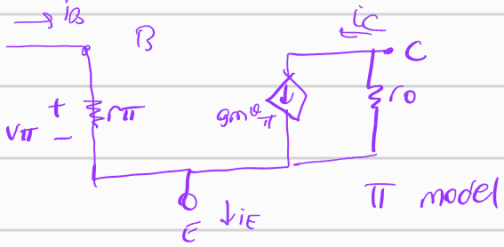




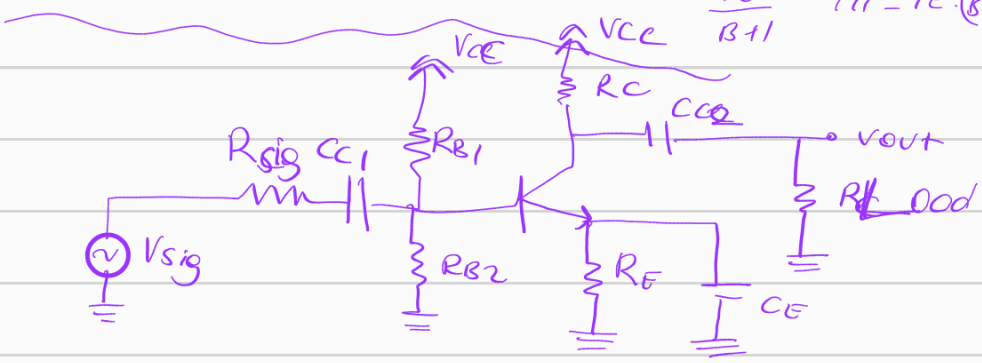
$r_o \rightarrow \frac{V_A}{I_C}$ $\quad g_m \rightarrow \frac{I_C}{V_T}$ $\quad i_B = \frac{V_\pi}{r_\pi}$ $\quad i_B = \frac{i_C}{\beta}$

$r_e = \frac{V_T}{I_E} = 1/g_m$ $\quad i_B + i_C = i_E$ $\quad r_E = \frac{\beta+1}{\beta} \cdot i_C$

$\boxed{r_\pi = (\beta+1)\, r_e}$ $\quad (\beta+1)\, i_B = i_E$

$r_\pi (i_B) = i_E \cdot r_e$

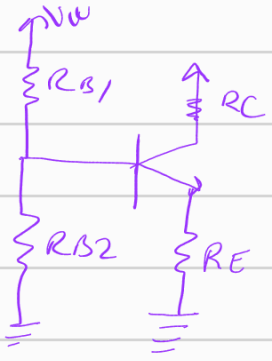
$\frac{i_E}{\beta+1}$ $\quad r_\pi = r_e \cdot (\beta+1)$

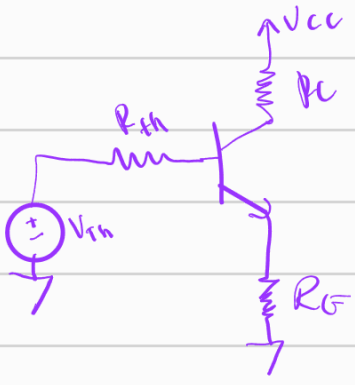


C_{C1}, C_{C2}, C_E high values

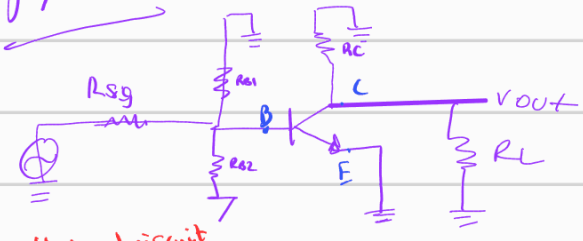
$A_v = \frac{v_o}{v_i} = ?$

DC

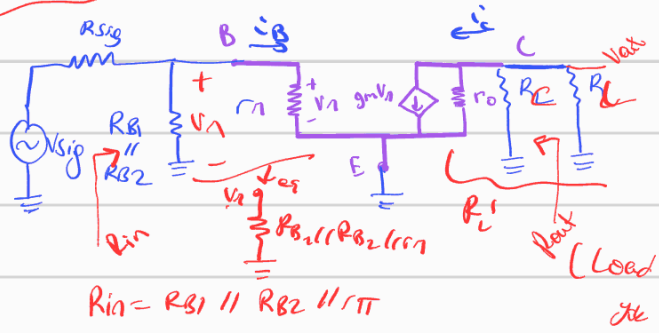




AC



Small signal circuit



$R_{in} = R_{B1} \,//\, R_{B2} \,//\, r_\pi$

$R_{out} = R_C // r_o$

$A_v = \frac{v_{out}}{v_{in}} = \frac{v_{out}}{v_{sig}} =$



$v_{out} = -R_L' \cdot v_{in} g_m$

I_1 sabit

$I_{R_o} + I_{R_L} = I_1$

$R_o \downarrow \uparrow \downarrow$

$v_{in} = \left(\frac{R_{B1} // R_{B2} // r_\pi}{r_\pi // R_{B1} // R_{B2} + R_{sig}} \right) v_{sig}$

$v_{out} = -g_m R_L' \cdot \left(\frac{R_{B1} // R_{B2} // r_\pi}{r_\pi // R_{B1} // R_{B2} + R_{sig}} \right) \cancel{v_{sig}}$

$v_{in} = \cancel{v_{sig}}$

$A_v = -g_m R_L' \left(\frac{R_{B1} // R_{B2} // r_\pi}{r_\pi // R_{B1} // R_{B2} + R_{sig}} \right)$

if $R_{sig} \approx 0 \Rightarrow A_v = -g_m R_L'$

Properties

High voltage gain

→ Moderate Input Resistance ⇒ $R_{B1} // R_{B2} // r_\pi$

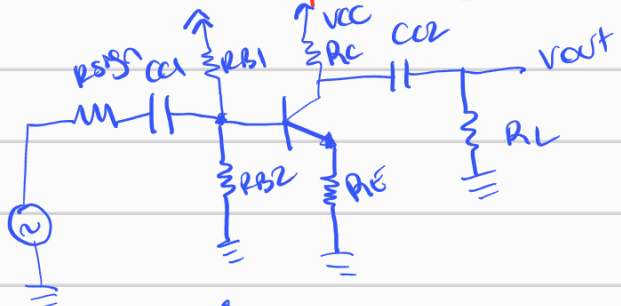
→ " High output Res → $R_C // r_o$

Rin büyük olsun isteriz ki

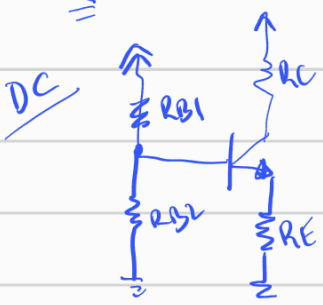
sinyal zayıflamasın girişte voltaj

düşümünü en aza indirgemek için

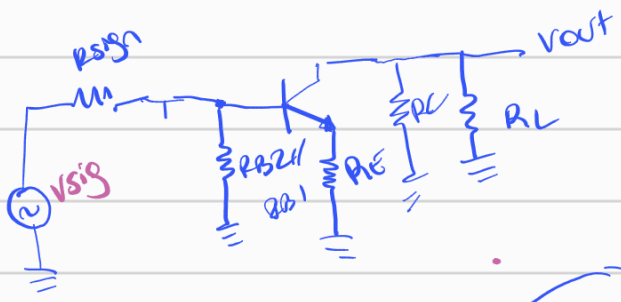
Common Emitter Amp. with Re

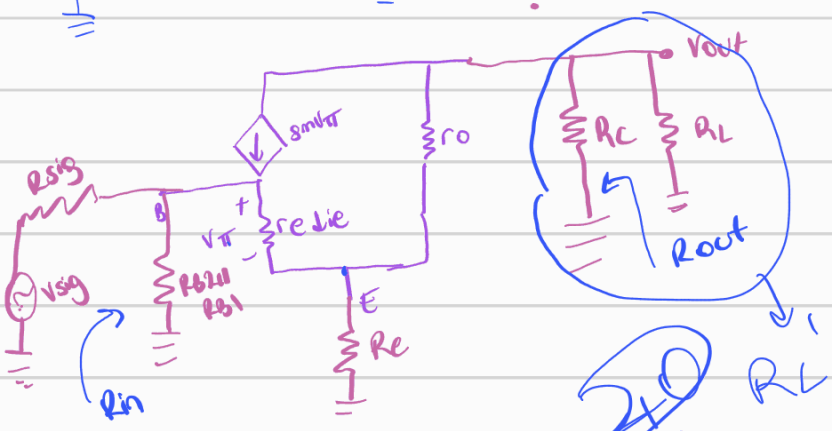


AC



AC

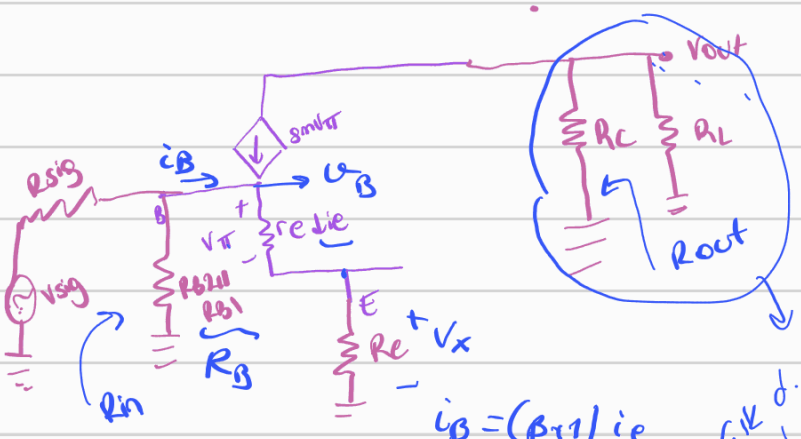




$R_{in} = R_{B2} \,//\, R_{B1} \,//\, ((\beta+1)(R_e + r_e))$

$R_{out} = R_C \,//\, (r_o + R_e)$

$V_A = \infty, \; r_o = \infty$



$i_B = (\beta+1)\, i_e$

$R_{in} = R_B \,//\, ((\beta+1)(r_e + R_e))$

$R_{out} = R_C$

akım kaynağı gibi davranır

voltaj kaynağı gibi davranır

$A_0 = \dfrac{v_{out}}{v_{in}} =$

$v_B = v_\pi + v_x$

$v_B = v_\pi + i_e R_e$

$v_{out} = -g_m v_\pi (R_C \,//\, R_L) \quad v_B = v_\pi + ((i_B + g_m v_\pi) R_e)$

$R_{sig} = 0$

$v_{sig} = v_b$

$$v_B = v_1 + i_B R_E + \frac{v_1 R_E}{R_E}$$

$$= 2v_1$$

$$v_x = i_E R_E$$

$$= (g_m v_1 + i_B) R_E$$

$$i_B = \frac{v_1}{r_1} \quad = \left(g_m v_1 + \frac{v_1}{r_1}\right) R_E$$

$$= \left(\frac{v_1}{R_E} + \frac{v_1}{r_1}\right) R_E$$

$$= \boxed{\frac{v_1}{r_e} \cdot R_E}$$

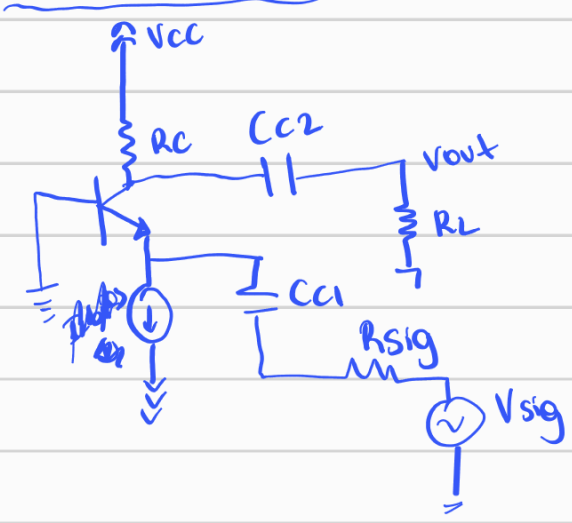
$$\frac{v_{out}}{v_{in}} = \frac{-g_m v_1 (R_C \parallel R_L)}{v_1 \left(1 + \frac{R_E}{r_e}\right)}$$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m (R_C \parallel R_L)}{r_e \left(1 + \frac{R_E}{r_e}\right)}$$

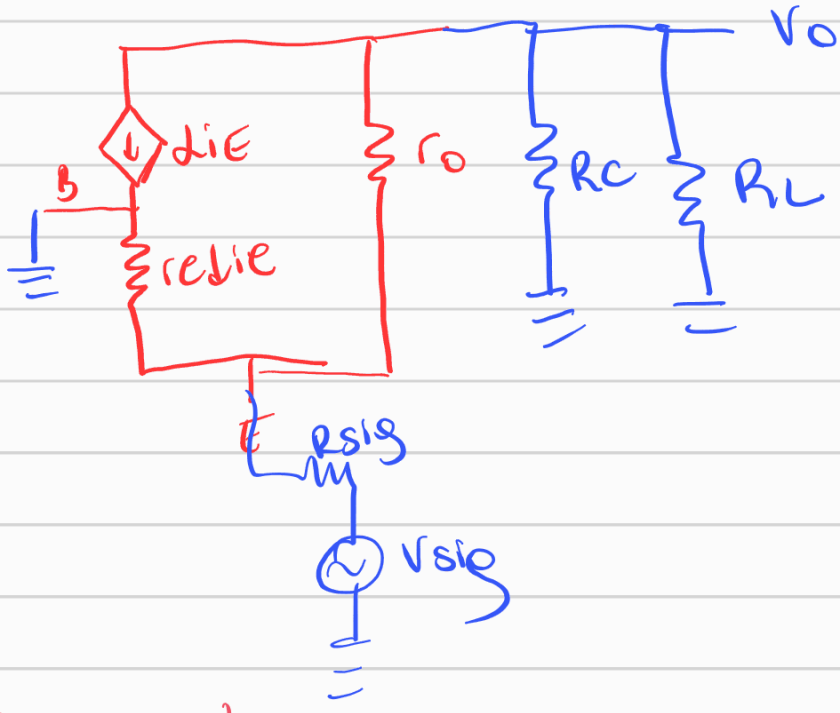
$$\approx -\frac{(R_C \parallel R_L)}{r_e + R_E}$$

$$A_v \approx -\frac{R_C \parallel R_L}{R_E + r_e}$$

Common-Base Amplifier

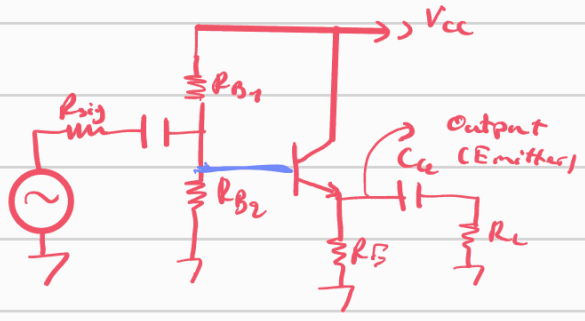


AC

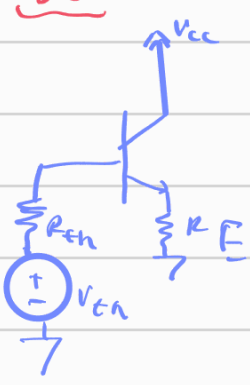


(Emitter-Follower)

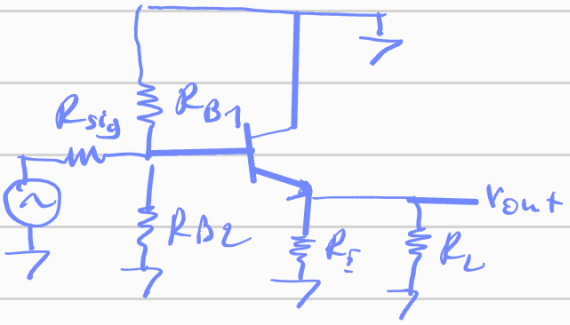
Common-Collector Amplifier

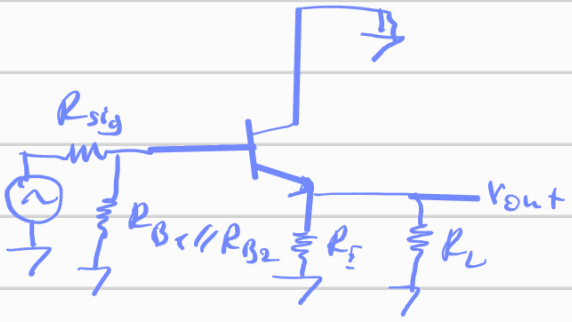


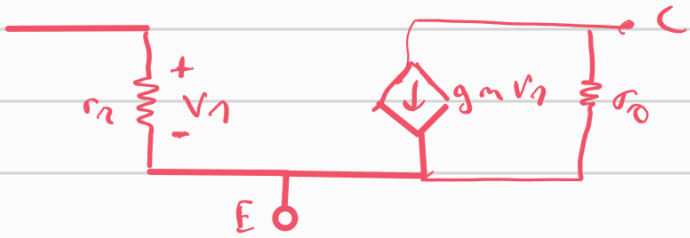
DC



AC







$V = IR$

$$\left(g_m V_\pi \cdot \quad + \frac{V_\pi}{r_\pi} \cdot \right) R_L'$$

$$= R_L' \cdot V_\pi \left(g_m + \frac{1}{r_\pi} \right)$$